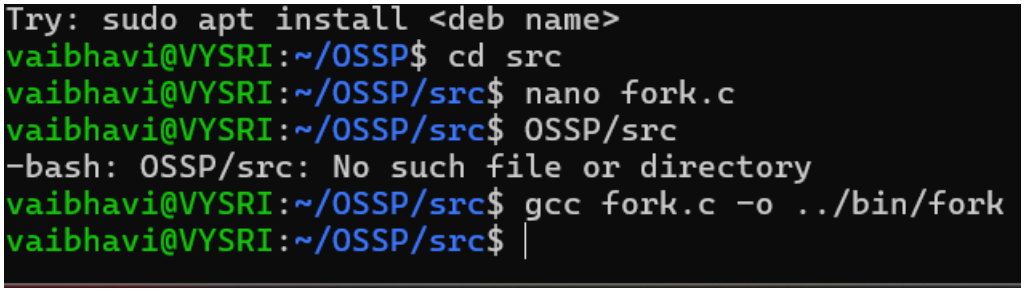
Name: S. Vaibhava Sree
Roll No: 2520030398

1. To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing




vaibhavi@VYSRI: ~/OSSP/src
vaibhavi@VYSRI:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
vaibhavi@VYSRI:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
26 packages can be upgraded. Run 'apt list --upgradable' to see them.
vaibhavi@VYSRI:~$ sudo apt install gcc make git
gcc is already the newest version (4:15.2.0-5ubuntu1).
make is already the newest version (4.4.1-3).
make set to manually installed.
git is already the newest version (1:2.53.0-1ubuntu1).
git set to manually installed.
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 26
vaibhavi@VYSRI:~$ mkdir OSSP
mkdir: OSSP: File exists
vaibhavi@VYSRI:~$ cd OSSP
vaibhavi@VYSRI:~/OSSP$ pwd
/home/vaibhavi/OSSP
vaibhavi@VYSRI:~/OSSP$ mkdir src include bin
mkdir: src: File exists
mkdir: include: File exists
mkdir: bin: File exists
vaibhavi@VYSRI:~/OSSP$ ls
Practical6 bin include memory_layout.c.save signal
allocation cow memory_layout pipe signal.c
allocation.c cow.c memory_layout.c pipe.c src
vaibhavi@VYSRI:~/OSSP$ bin include src
Command 'bin' not found, did you mean:
command 'din' from deb din (58.1+ds-1build2)
command 'bip' from deb bip (0.9.3-1.2)
command 'ben' from deb ben (1.15ubuntu3)
command 'win' from deb wily (0.13.42-4build1)
command 'tin' from deb tin (1:2.6.6~20260126-1)


Try: sudo apt install <deb name>
vaibhavi@VYSRI:~/OSSP$ cd src
vaibhavi@VYSRI:~/OSSP/src$ nano fork.c
vaibhavi@VYSRI:~/OSSP/src$ OSSP/src
-bash: OSSP/src: No such file or directory
vaibhavi@VYSRI:~/OSSP/src$ gcc fork.c -o ../bin/fork
vaibhavi@VYSRI:~/OSSP/src$


vaibhavi@VYSRI: ~/OSSP/src
GNU nano 8.7.1 process.c
#include <stdio.h>
#include <unistd.h>
int main() {
printf("Process ID: %d\n", getpid());
printf("Parent Process ID: %d\n", getppid());
return 0;
}
[Read 9 lines]
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify




```
vaibhavi@VYSRI:~/OSSP/src$ nano fork.c
vaibhavi@VYSRI:~/OSSP/src$ OSSP/src
-bash: OSSP/src: No such file or directory
vaibhavi@VYSRI:~/OSSP/src$ gcc fork.c -o ../bin/fork
vaibhavi@VYSRI:~/OSSP/src$ ../bin/fork
Before fork()
After fork()
After fork()
```

```
vaibhavi@VYSRI:~/OSSP/src$ nano process.c
vaibhavi@VYSRI:~/OSSP/src$ gcc process.c -o ../bin/process
vaibhavi@VYSRI:~/OSSP/src$ ../bin/process
Process ID: 903
Parent Process ID: 324
```

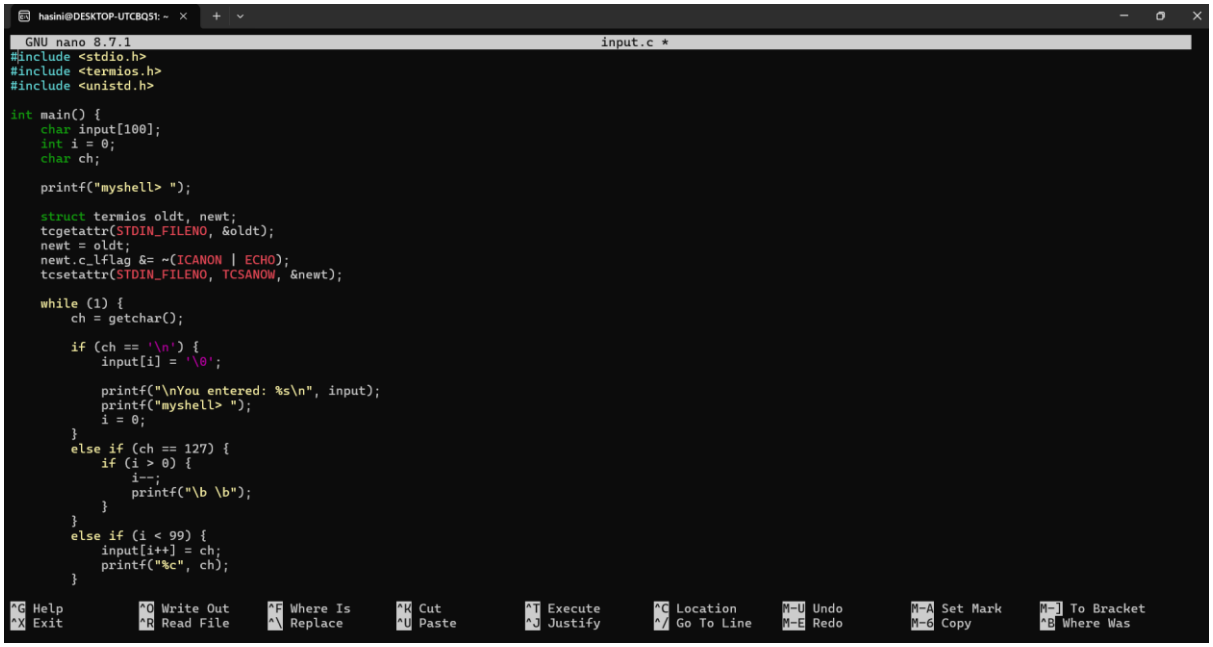
```
vaibhavi@VYSRI:~/OSSP/src$ pstree -p
systemd(1)─┬─agetty(277)
           ├─chronyd-starter(151)───chronyd(263)───chronyd(265)
           ├─cron(152)
           ├─dbus-daemon(153)
           ├─init-systemd(Ub(2)─┬─SessionLeader(322)───Relay(324)(323)───ba+
           │                    ├─init(6)───{init}(7)
           │                    ├─login(325)───bash(399)
           │                    └─{init-systemd(Ub}(8)
           ├─networkd-dispat(157)
           ├─polkitd(560)─┬─{polkitd}(562)
           │              ├─{polkitd}(563)
           │              └─{polkitd}(564)
           ├─rsyslogd(210)─┬─{rsyslogd}(258)
           │               ├─{rsyslogd}(259)
           │               └─{rsyslogd}(260)
           ├─systemd(369)───(sd-pam)(371)
           ├─systemd-journal(46)
           ├─systemd-logind(162)
           ├─systemd-resolve(83)
           ├─systemd-udevd(89)
           └─unattended-upgr(273)───{unattended-upgr}(284)
```

```
vaibhavi@VYSRI:~/OSSP/src$ strace -V
strace -- version 6.19
Copyright (c) 1991-2026 The strace developers <https://strace.io>.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

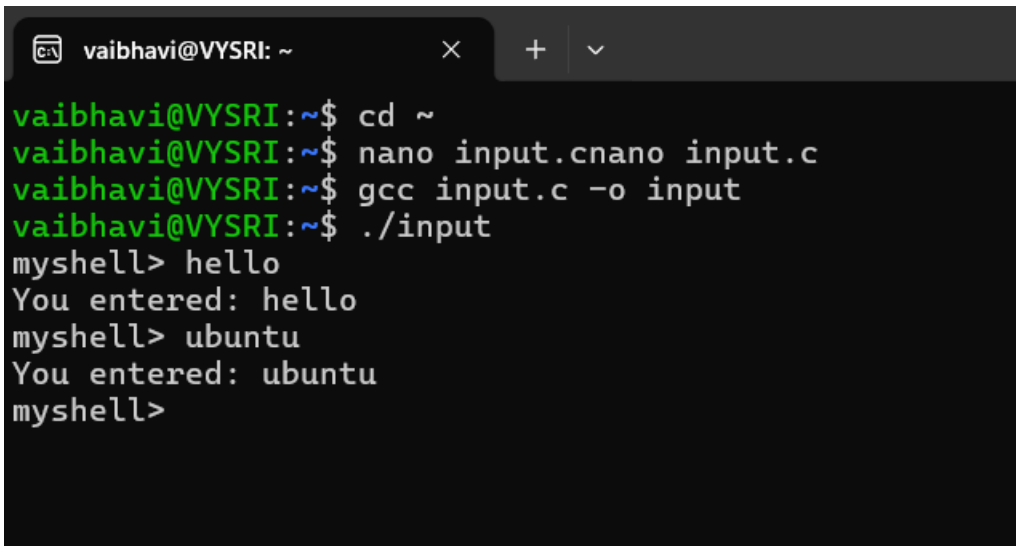
Optional features enabled: stack-trace=libunwind stack-demangle m32-mpers mx
32-mpers
vaibhavi@VYSRI:~/OSSP/src$ sudo apt install strace
strace is already the newest version (6.19+ds-0ubuntu5).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 26
vaibhavi@VYSRI:~/OSSP/src$ strace ../bin/process
execve("../bin/process", ["../bin/process"], 0x7ffe12f72ed0 /* 27 vars */) =
 0
brk(NULL)                               = 0x58608939d000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0
x74fee47d4000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directo
ry)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20815, ...}) = 0
mmap(NULL, 20815, PROT_READ, MAP_PRIVATE, 3, 0) = 0x74fee47ce000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC)
= 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0\0"...
, 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..
., 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2190608, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..
., 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x74fee440
0000
mmap(0x74fee4428000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP
_DENYWRITE, 3, 0x28000) = 0x74fee4428000
mmap(0x74fee45c0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
 3, 0x1c0000) = 0x74fee45c0000
```

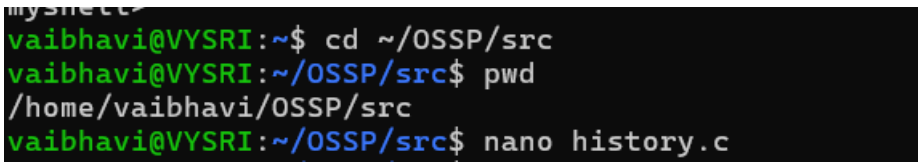
```
ANONYMOUS, -1, 0) = 0x74fee4614000
close(3)                                = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x74fee47cb000
arch_prctl(ARCH_SET_FS, 0x74fee47cb740) = 0
set_tid_address(0x74fee47cbd68)         = 1090
set_robust_list(0x74fee47cba20, 24)     = 0
rseq(0x74fee47cb6a0, 0x20, 0, 0x53053053) = 0
mprotect(0x74fee460e000, 16384, PROT_READ) = 0
mprotect(0x58604f6e9000, 4096, PROT_READ) = 0
mprotect(0x74fee4819000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINI
TY}) = 0
getrandom("\x76\xa8\x10\x3a\x3d\x78\xf3\x66", 8, GRND_NONBLOCK) = 8
munmap(0x74fee47ce000, 20815)           = 0
getpid()                                = 1090
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
brk(NULL)                               = 0x58608939d000
brk(0x5860893be000)                     = 0x5860893be000
write(1, "Process ID: 1090\n", 17Process ID: 1090
)       = 17
getppid()                               = 1088
write(1, "Parent Process ID: 1088\n", 24Parent Process ID: 1088
) = 24
exit_group(0)                           = ?
+++ exited with 0 +++
vaibhavi@VYSRI:~/OSSP/src$
```

2.




hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1
input.c *
char ch;
printf("myshell> ");
struct termios oldt, newt;
tcgetattr(STDIN_FILENO, &oldt);
newt = oldt;
newt.c_lflag &= ~(ICANON | ECHO);
tcsetattr(STDIN_FILENO, TCSANOW, &newt);
while (1) {
ch = getchar();
if (ch == '\n') {
input[i] = '\0';
printf("\nYou entered: %s\n", input);
printf("myshell> ");
i = 0;
}
else if (ch == 127) {
if (i > 0) {
i--;
printf("\b \b");
}
}
else if (i < 99) {
input[i++] = ch;
printf("%c", ch);
}
}
tcsetattr(STDIN_FILENO, TCSANOW, &oldt);
return 0;
}
^G Help
^X Exit
^O Write Out
^R Read File
^F Where Is
^\ Replace
^K Cut
^U Paste
^T Execute
^J Justify
^C Location
^/ Go To Line
M-U Undo
M-E Redo
M-A Set Mark
M-6 Copy
M-] To Bracket
^B Where Was


vaibhavi@VYSRI: ~
vaibhavi@VYSRI:~$ cd ~
vaibhavi@VYSRI:~$ nano input.cnano input.c
vaibhavi@VYSRI:~$ gcc input.c -o input
vaibhavi@VYSRI:~$ ./input
myshell> hello
You entered: hello
myshell> ubuntu
You entered: ubuntu
myshell>

3.

vaibhavi@VYSRI:~$ cd ~/OSSP/src
vaibhavi@VYSRI:~/OSSP/src$ pwd
/home/vaibhavi/OSSP/src
vaibhavi@VYSRI:~/OSSP/src$ nano history.c

vaibhavi@VYSRI: ~/OSSP/src

GNU nano 8.7.1 history.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <termios.h>
#include <unistd.h>

#define MAX_HISTORY 10

char *history[MAX_HISTORY];
int history_count = 0;
int history_pos = 0;

void add_history(char *command) {
    if (strlen(command) == 0)
        return;

    if (history_count < MAX_HISTORY) {
        history[history_count] = malloc(strlen(command) + 1);
        strcpy(history[history_count], command);
        history_count++;
    }
}

void show_history_command() {
    if (history_pos >= 0 && history_pos < history_count)
        printf("\r\033[Kmyshell> %s", history[history_pos]);
}

int main() {
    char input[100];
    int i = 0;
    char ch;

    struct termios oldt, newt;

    tcgetattr(STDIN_FILENO, &oldt);
    newt = oldt;
    newt.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    while (1) {
        printf("\nmyshell> ");
        fflush(stdout);

        i = 0;

        while (1) {
            ch = getchar();

            if (ch == '\n') {
                input[i] = '\0';
```

vaibhavi@VYSRI: ~/OSSP/src

GNU nano 8.7.1 history.c *

```
                input[i] = '\0';

                if (strcmp(input, "exit") == 0)
                    break;

                printf("\nYou entered: %s\n", input);

                add_history(input);
                history_pos = history_count;

                break;
            }

            else if (ch == 127) {
                if (i > 0) {
                    i--;
                    printf("\b \b");
                    fflush(stdout);
                }
            }

            else if (ch == 27) {
                char ch1 = getchar();
                char ch2 = getchar();

                if (ch1 == '[' && ch2 == 'A') {
                    if (history_count > 0 && history_pos > 0) {
                        history_pos--;
                        show_history_command();

                        strcpy(input, history[history_pos]);
                        i = strlen(input);
                    }
                }

                else if (ch1 == '[' && ch2 == 'B') {
                    if (history_pos < history_count - 1) {
                        history_pos++;
                        show_history_command();

                        strcpy(input, history[history_pos]);
                        i = strlen(input);
                    }
                }

                fflush(stdout);
            }

            else if (i < 99) {
                input[i++] = ch;
                printf("%c", ch);
                fflush(stdout);
            }
        }

        if (strcmp(input, "exit") == 0)
            break;
    }

    tcsetattr(STDIN_FILENO, TCSANOW, &oldt);

    for (int j = 0; j < history_count; j++)
       free(history[j]);

    return 0;
}
```

```
vaibhavi@VYSRI:~/OSSP/src$ gcc history.c -o history
vaibhavi@VYSRI:~/OSSP/src$ ./history

myshell> hello
You entered: hello

myshell> ubuntu
You entered: ubuntu

myshell> linux
You entered: linux

myshell>
```

hasini@DESKTOP-UTCBQ51: ~

```
 GNU nano 8.7.1                          memory.c *
#include <stdlib.h>
#include <string.h>

struct Node {
    char data[50];
    struct Node *next;
};

int main() {
    struct Node *head = NULL;
    struct Node *newNode;
    struct Node *temp;

    // Allocate memory
    newNode = malloc(sizeof(struct Node));

    if (newNode == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    strcpy(newNode->data, "Hello");
    newNode->next = NULL;
    head = newNode;

    printf("Data: %s\n", head->data);

    // Release memory
    temp = head;
    free(temp);
    head = NULL;

    printf("Memory released successfully\n");

    return 0;
}

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

```
vaibhavi@VYSRI:~/OSSP/src$ nano memory.c
vaibhavi@VYSRI:~/OSSP/src$ gcc memory.c -o memory
vaibhavi@VYSRI:~/OSSP/src$ ./memory
Data: Hello
Memory released successfully
```

vaibhavi@VYSRI: ~/OSSP/src

GNU nano 8.7.1 input.c *

```c
#include <stdio.h>
#include <termios.h>
#include <unistd.h>

int main() {
    char input[100];
    int i = 0;
    char ch;

    printf("myshell> ");

    struct termios oldt, newt;
    tcgetattr(STDIN_FILENO, &oldt);
    newt = oldt;
    newt.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    while (1) {
        ch = getchar();

        if (ch == '\n') {
            input[i] = '\0';

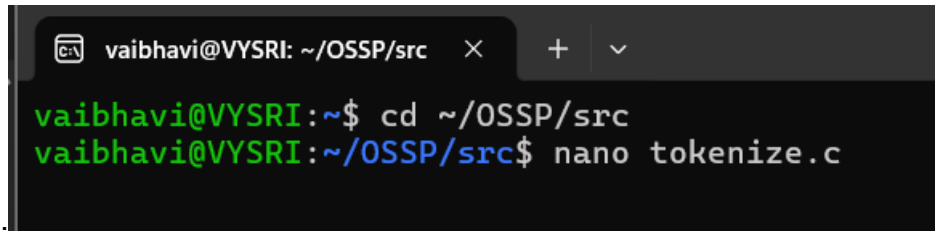
            printf("\nYou entered: %s\n", input);
            printf("myshell> ");
            i = 0;
        }
        else if (ch == 127) {
            if (i > 0) {
                i--;
                printf("\b \b");
            }
        }
        else if (i < 99) {
            input[i++] = ch;
            printf("%c", ch);
        }
    }

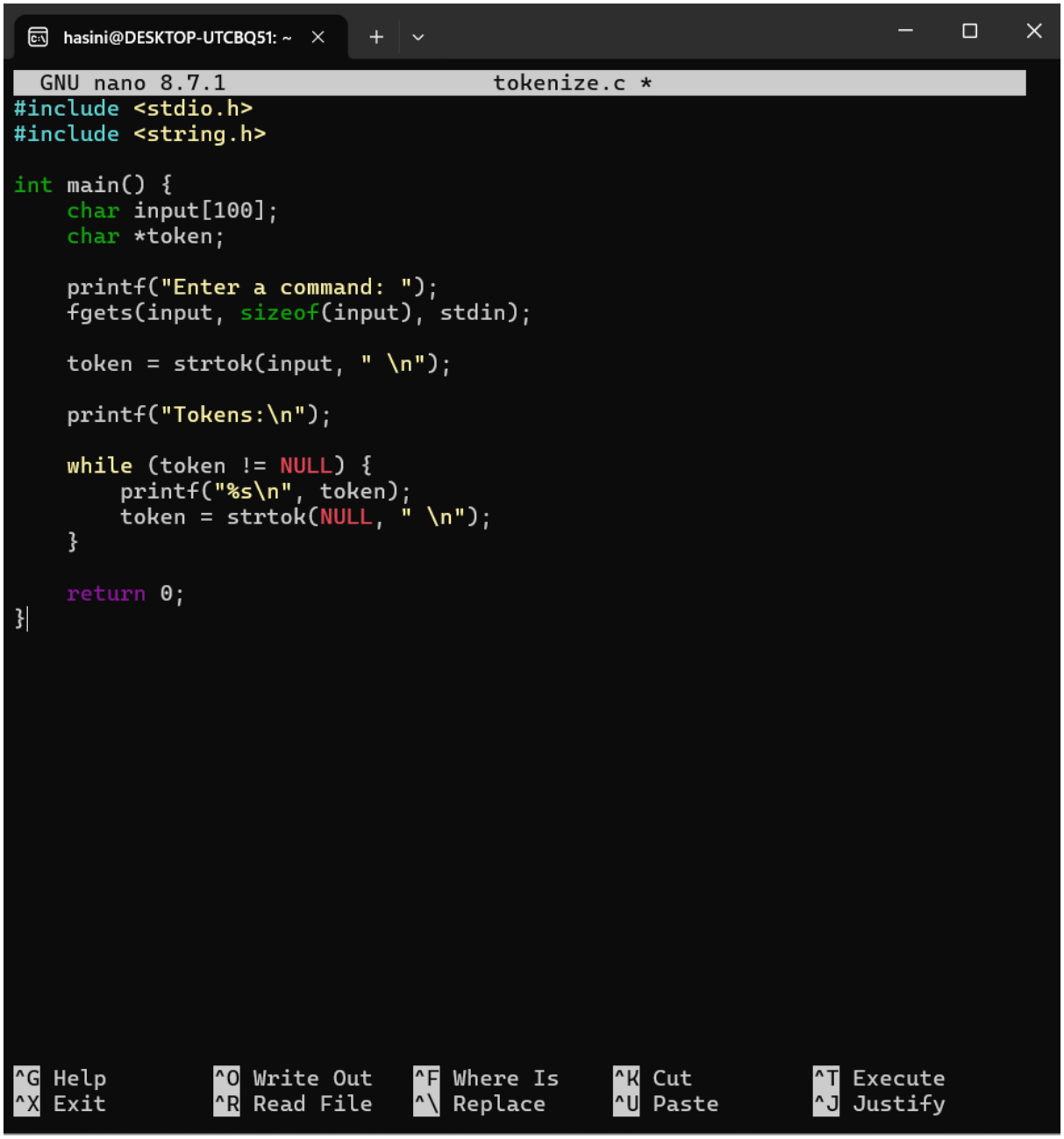
    tcsetattr(STDIN_FILENO, TCSANOW, &oldt);

    return 0;
}
```

```
vaibhavi@VYSRI:~/OSSP/src$ nano input.c
vaibhavi@VYSRI:~/OSSP/src$ gcc input.c -o input
vaibhavi@VYSRI:~/OSSP/src$ ./input
myshell> hi
You entered: hi
myshell> linux
You entered: linux
myshell>
```

4.





```
vaibhavi@VYSRI:~/OSSP/src$ ../bin/tokenize
Enter a command: ls -l file.txt
Tokens:
ls
-l
file.txt
vaibhavi@VYSRI:~/OSSP/src$
```

5.

```
vaibhavi@VYSRI:~/OSSP/src$ nano parse.c
vaibhavi@VYSRI:~/OSSP/src$ gcc parse.c -o ../bin/parse
vaibhavi@VYSRI:~/OSSP/src$ ../bin/parse
Enter command: ls file.txt
Command: ls
Argument: file.txt
vaibhavi@VYSRI:~/OSSP/src$
```



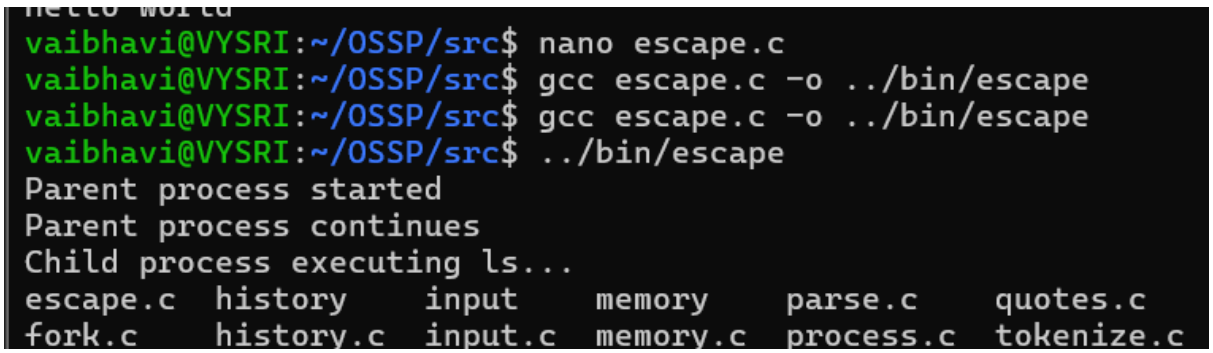
6.

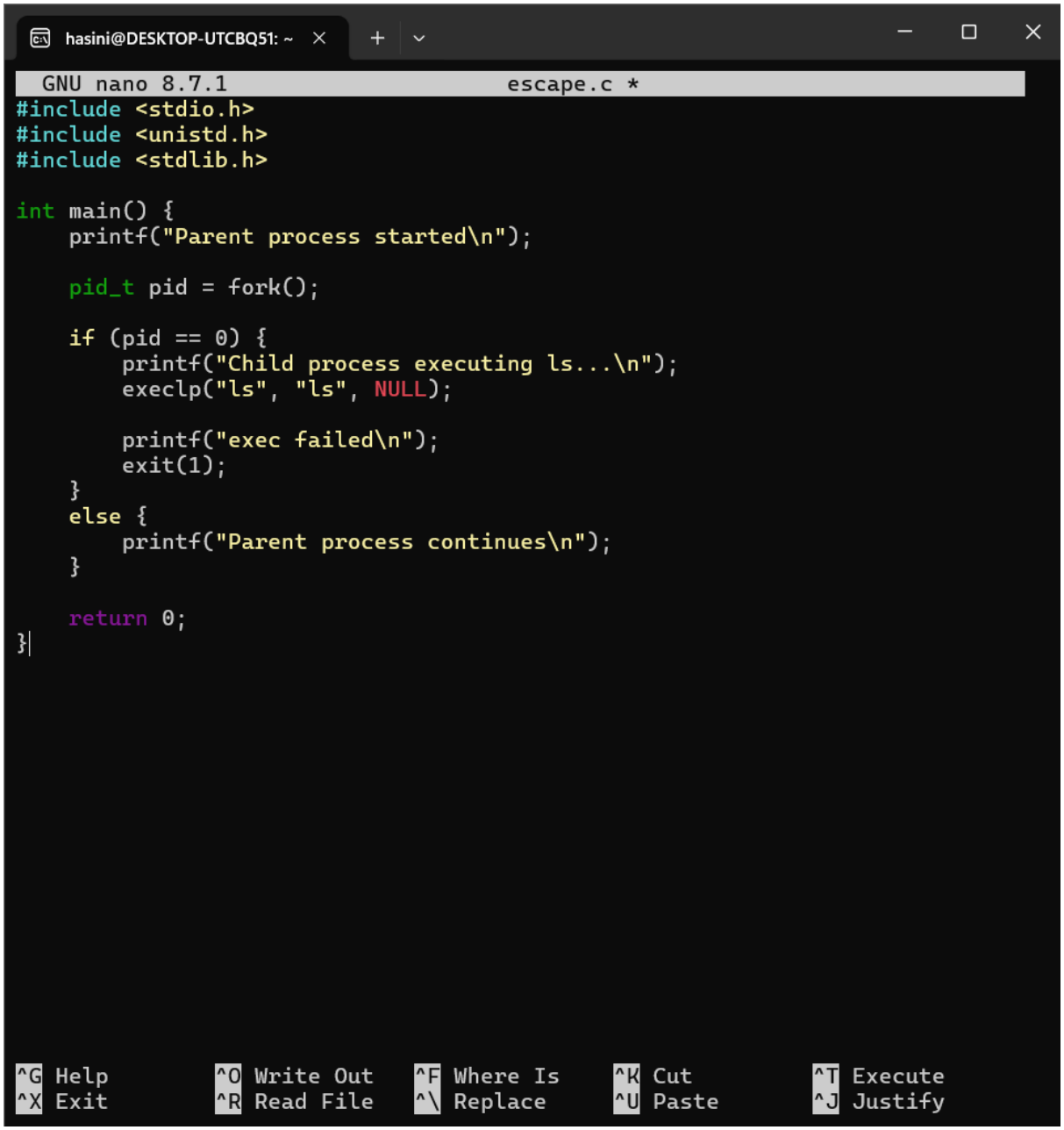
```
vaibhavi@VYSRI:~/OSSP/src$ nano quotes.c
vaibhavi@VYSRI:~/OSSP/src$ gcc quotes.c -o ../bin/quotes
vaibhavi@VYSRI:~/OSSP/src$ ../bin/quotes
Enter a command: "Hello World"
Tokens:
Hello World
```


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 quotes.c *
#include <stdio.h>
#include <string.h>
int main()
{
char input[200];
char token[100];
int i = 0;
int j = 0;
char quote = 0;
printf("Enter a command: ");
fgets(input, sizeof(input), stdin);
printf("Tokens:\n");
while (input[i] != '\0' && input[i] != '\n')
{
if (input[i] == '"' || input[i] == '\'')
{
if (quote == 0)
{
quote = input[i];
}
else if (quote == input[i])
{
quote = 0;
}
else
{
token[j++] = input[i];
}
}
else if (input[i] == ' ' && quote == 0)
{
if (j > 0)
{
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 quotes.c *
quote = input[i];
}
else if (quote == input[i])
{
quote = 0;
}
else
{
token[j++] = input[i];
}
}
else if (input[i] == ' ' && quote == 0)
{
if (j > 0)
{
token[j] = '\0';
printf("%s\n", token);
j = 0;
}
}
else
{
token[j++] = input[i];
}
i++;
}
if (j > 0)
{
token[j] = '\0';
printf("%s\n", token);
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

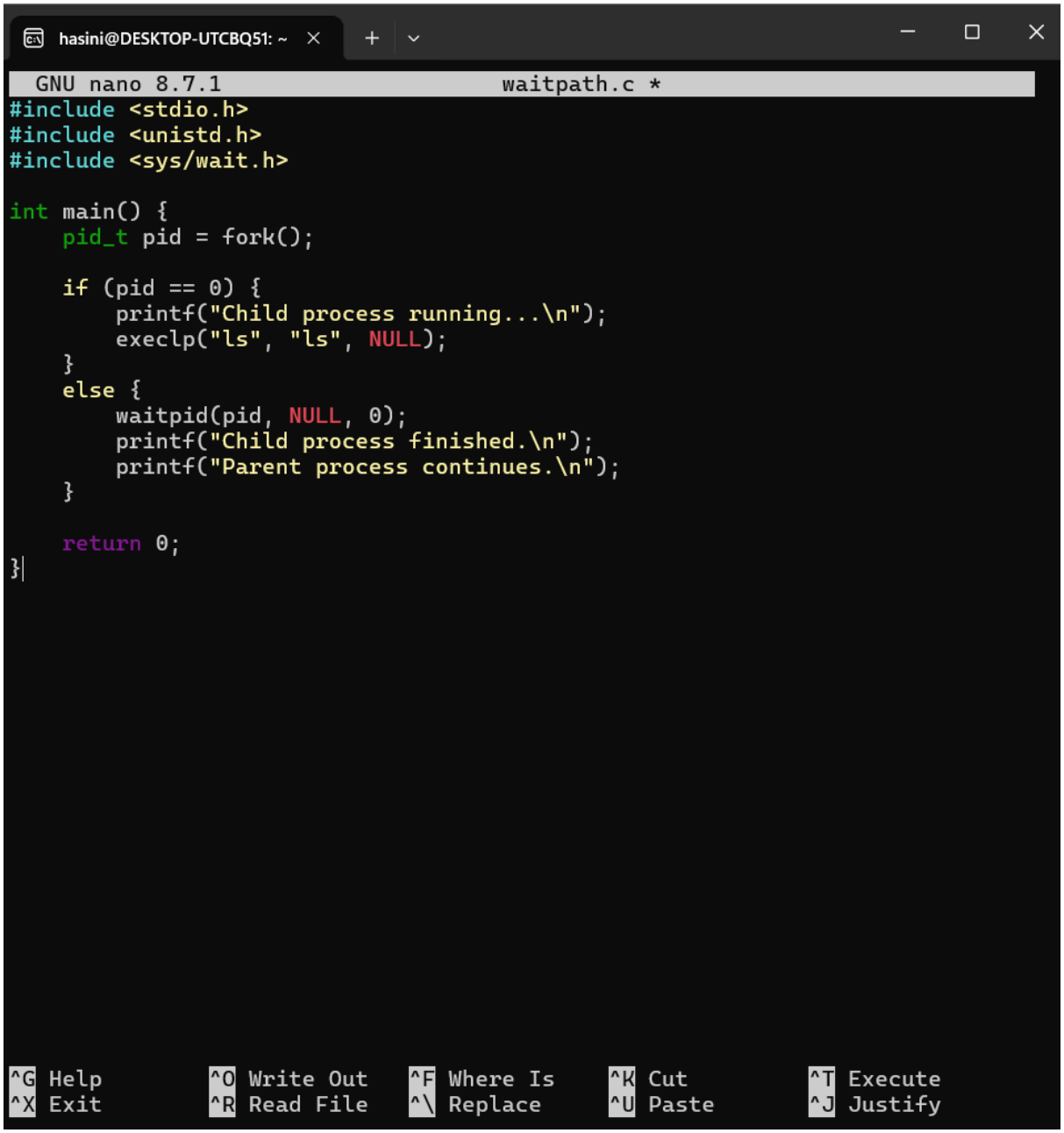
7.


vaibhavi@VYSRI:~/OSSP/src$ nano escape.c
vaibhavi@VYSRI:~/OSSP/src$ gcc escape.c -o ../bin/escape
vaibhavi@VYSRI:~/OSSP/src$ gcc escape.c -o ../bin/escape
vaibhavi@VYSRI:~/OSSP/src$ ../bin/escape
Parent process started
Parent process continues
Child process executing ls...
escape.c history input memory parse.c quotes.c
fork.c history.c input.c memory.c process.c tokenize.c



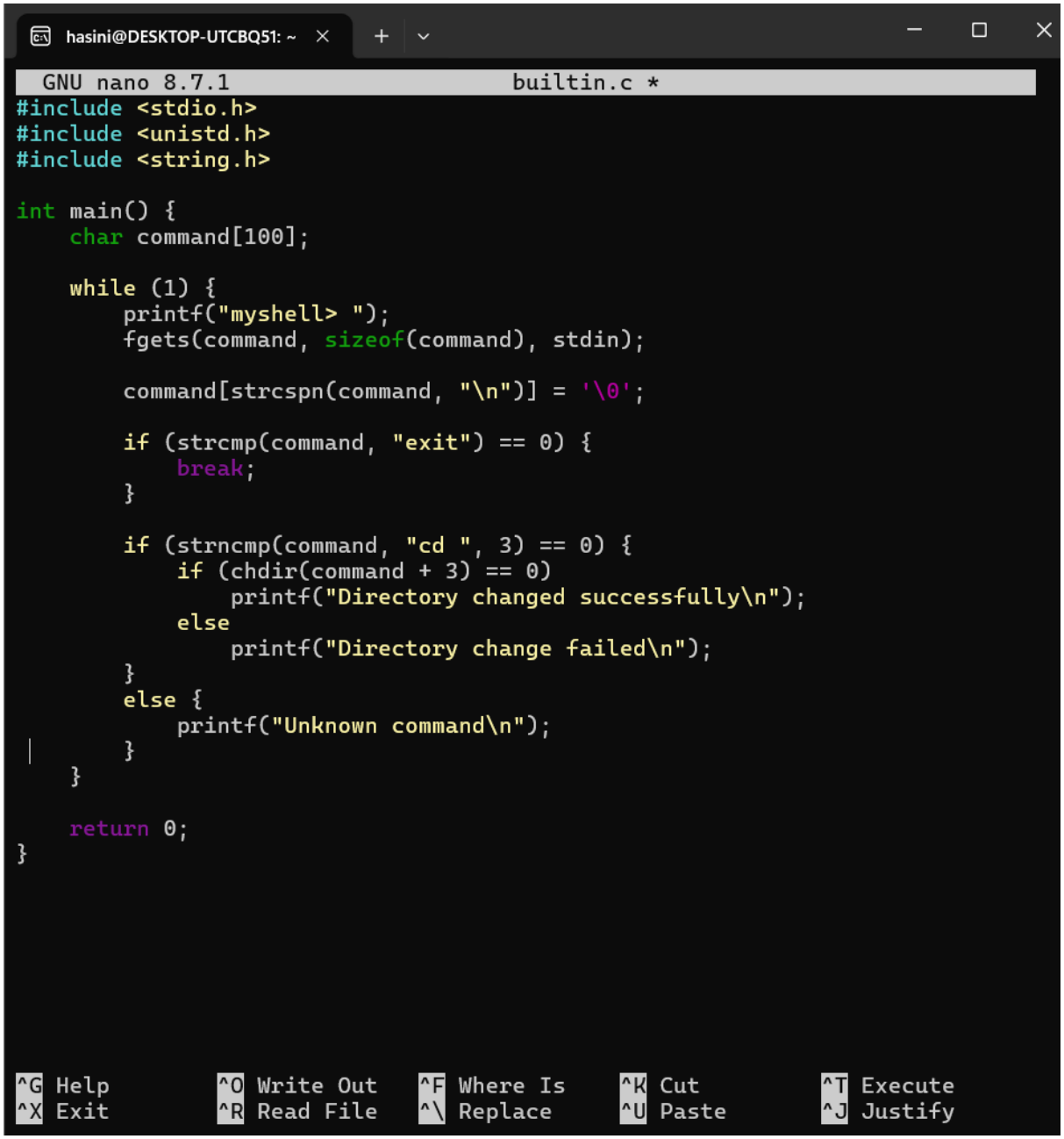
8.

```
vaibhavi@VYSRI:~/OSSP/src$ nano waitpath.c
vaibhavi@VYSRI:~/OSSP/src$ gcc waitpath.c -o ../bin/waitpath
vaibhavi@VYSRI:~/OSSP/src$ ../bin/waitpath
Child process running...
escape.c  history    input    memory    parse.c    quotes.c    waitpath.c
fork.c    history.c  input.c  memory.c  process.c  tokenize.c
Child process finished.
Parent process continues.
vaibhavi@VYSRI:~/OSSP/src$
```



9.

```
vaibhavi@VYSRI:~/OSSP/src$ cd ~/OSSP/src
vaibhavi@VYSRI:~/OSSP/src$ nano builtin.c
vaibhavi@VYSRI:~/OSSP/src$ gcc builtin.c -o ../bin/builtin
vaibhavi@VYSRI:~/OSSP/src$ ../bin/builtin
myshell> cd /tmp
Directory changed successfully
myshell> hello
Unknown command
myshell> exit
vaibhavi@VYSRI:~/OSSP/src$
```


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 builtin.c *
#include <stdio.h>
#include <unistd.h>
#include <string.h>
int main() {
char command[100];
while (1) {
printf("myshell> ");
fgets(command, sizeof(command), stdin);
command[strcspn(command, "\n")] = '\0';
if (strcmp(command, "exit") == 0) {
break;
}
if (strncmp(command, "cd ", 3) == 0) {
if (chdir(command + 3) == 0)
printf("Directory changed successfully\n");
else
printf("Directory change failed\n");
}
else {
printf("Unknown command\n");
}
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

10.

```
  GNU nano 8.7.1                    builtins2.c *
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <stdlib.h>

int main() {
    char command[100];

    while (1) {
        printf("myshell> ");
        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "pwd") == 0) {
            char path[200];
            getcwd(path, sizeof(path));
            printf("%s\n", path);
        }
        else if (strcmp(command, "exit") == 0) {
            break;
        }
        else if (strncmp(command, "export ", 7) == 0) {
            char *value = command + 7;
            putenv(value);
            printf("Variable exported\n");
        }
        else {
            printf("Unknown command\n");
        }
    }

    return 0;
}

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

```
myshell> exit
vaibhavi@VYSRI:~/OSSP/src$ nano builtins2.c
vaibhavi@VYSRI:~/OSSP/src$ gcc builtins2.c -o ../bin/builtins2
vaibhavi@VYSRI:~/OSSP/src$ ../bin/builtins2
myshell> pwd
/home/vaibhavi/OSSP/src
myshell> export NAME=Vaibhavasree
Variable exported
myshell> hi
Unknown command
myshell>
```

11.

hasini@DESKTOP-UTCBQ51: ~

GNU nano 8.7.1 pipeline.c *

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int fd[2];
    pipe(fd);

    pid_t pid = fork();

    if (pid == 0) {
        close(fd[0]);
        dup2(fd[1], STDOUT_FILENO);
        close(fd[1]);

        execlp("ls", "ls", NULL);
    }
    else {
        close(fd[1]);
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);

        execlp("wc", "wc", "-l", NULL);

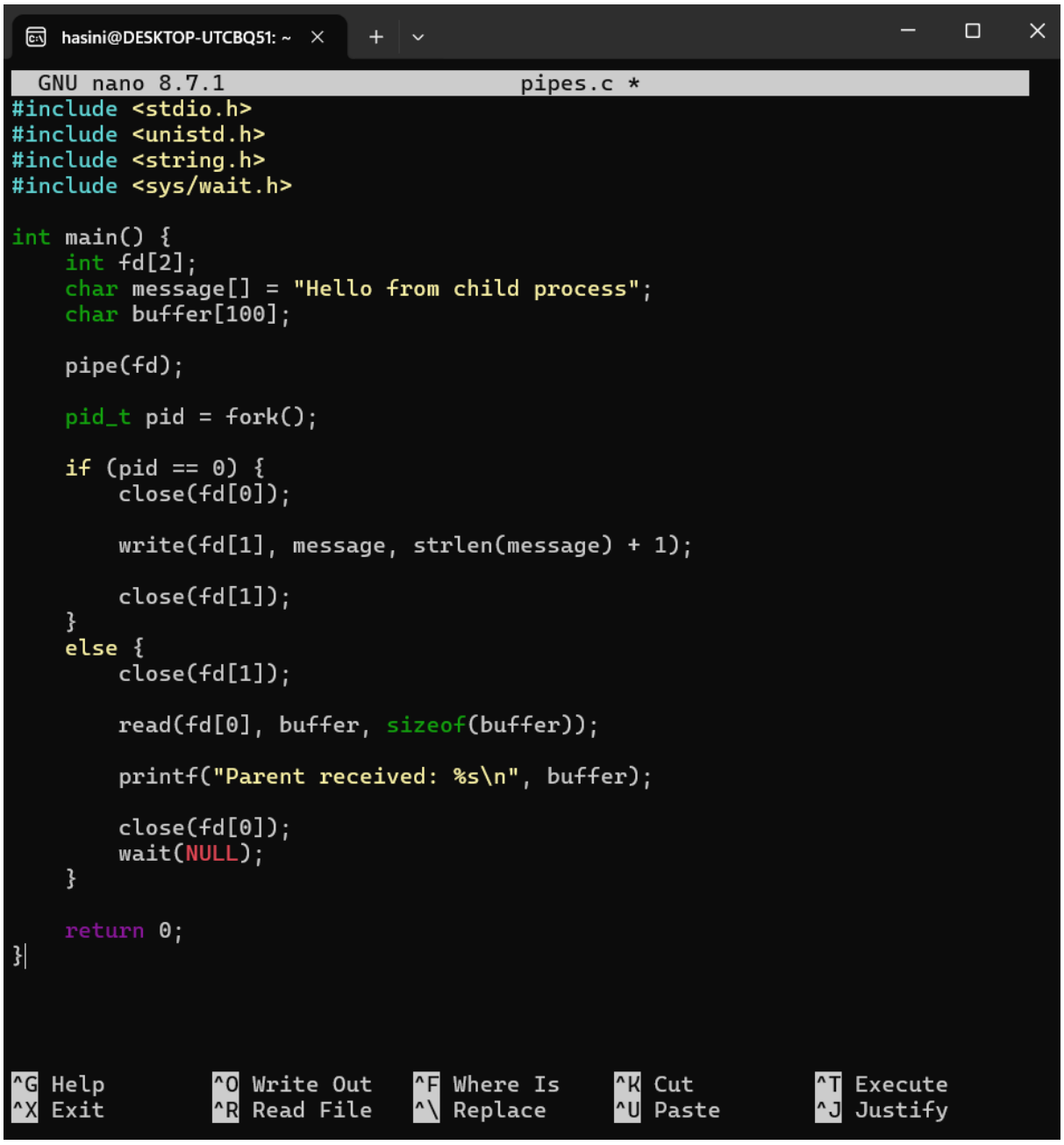
        wait(NULL);
    }

    return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

```
vaibhavi@VYSRI:~/OSSP/src$ nano pipeline.c
vaibhavi@VYSRI:~/OSSP/src$ gcc pipeline.c -o ../bin/pipeline
vaibhavi@VYSRI:~/OSSP/src$ ../bin/pipeline
16
vaibhavi@VYSRI:~/OSSP/src$
```

12.




```
vaibhavi@VYSRI:~/OSSP/src$ nano pipes.c
vaibhavi@VYSRI:~/OSSP/src$ gcc pipes.c -o ../bin/pipes
vaibhavi@VYSRI:~/OSSP/src$ ../bin/pipes
Parent received: Hello from child process
vaibhavi@VYSRI:~/OSSP/src$
```

13.

hasini@DESKTOP-UTCBQ51: ~

```
 GNU nano 8.7.1                          redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {
        perror("open");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("This text is redirected to a file.\n");

    return 0;
}

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

```
vaibhavi@VYSRI:~/OSSP/src$ nano redirect.c
vaibhavi@VYSRI:~/OSSP/src$ gcc redirect.c -o ../bin/redirect
vaibhavi@VYSRI:~/OSSP/src$ ../bin/redirect
vaibhavi@VYSRI:~/OSSP/src$ cat output.txt
This text is redirected to a file.
vaibhavi@VYSRI:~/OSSP/src$
```

14.

hasini@DESKTOP-UTCBQ51: ~

GNU nano 8.7.1 append.c *

```c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("output.txt", O_WRONLY | O_CREAT | O_APPEND, 0644);

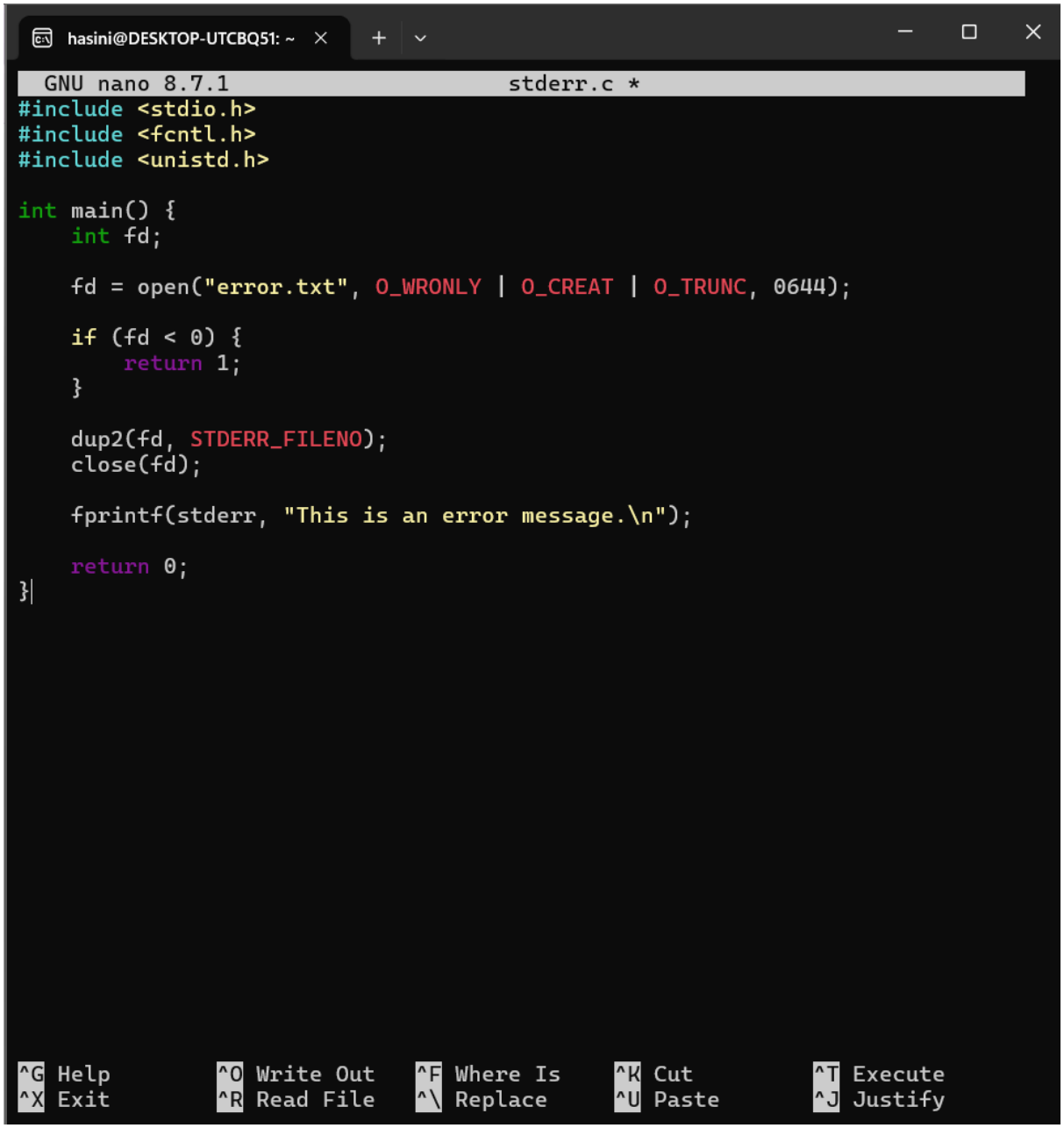
    if (fd < 0) {
        perror("open");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("This line is appended.\n");

    return 0;
}
```

```
vaibhavi@VYSRI:~/OSSP/src$ nano append.c
vaibhavi@VYSRI:~/OSSP/src$ gcc append.c -o ../bin/append
vaibhavi@VYSRI:~/OSSP/src$ ../bin/append
vaibhavi@VYSRI:~/OSSP/src$ cat output.txt
This text is redirected to a file.
This line is appended.
vaibhavi@VYSRI:~/OSSP/src$
```




```
vaibhavi@VYSRI:~/OSSP/src$ nano stderr.c
vaibhavi@VYSRI:~/OSSP/src$ gcc stderr.c -o ../bin/stderr
vaibhavi@VYSRI:~/OSSP/src$ ../bin/stderr
vaibhavi@VYSRI:~/OSSP/src$ cat error.txt
This is an error message.
vaibhavi@VYSRI:~/OSSP/src$
```

15.

vaibhavi@VYSRI: ~/OSSP/src

GNU nano 8.7.1 complex.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char input[200];

    printf("myshell> ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    if (strcmp(input, "exit") == 0)
        return 0;

    char *command1 = strtok(input, "|");
    char *command2 = strtok(NULL, "|");

    if (command2 != NULL)
    {
        int pipefd[2];
        pipe(pipefd);

        pid_t pid1 = fork();

        if (pid1 == 0)
        {
            dup2(pipefd[1], STDOUT_FILENO);
            close(pipefd[0]);
            close(pipefd[1]);

            char *args[10];
            int i = 0;
            char *token = strtok(command1, " ");
```

vaibhavi@VYSRI: ~/OSSP/src

GNU nano 8.7.1 comp

```
            char *token = strtok(command1, " ");

            while (token != NULL)
            {
                args[i++] = token;
                token = strtok(NULL, " ");
            }

            args[i] = NULL;
            execvp(args[0], args);
            perror("exec failed");
            exit(1);
        }

        pid_t pid2 = fork();

        if (pid2 == 0)
        {
            dup2(pipefd[0], STDIN_FILENO);
            close(pipefd[1]);
            close(pipefd[0]);

            char *args[10];
            int i = 0;
            char *token = strtok(command2, " ");

            while (token != NULL)
            {
                args[i++] = token;
                token = strtok(NULL, " ");
            }

            args[i] = NULL;
            execvp(args[0], args);
            perror("exec failed");
            exit(1);
        }

        close(pipefd[0]);
        close(pipefd[1]);

        waitpid(pid1, NULL, 0);
        waitpid(pid2, NULL, 0);
    }

    return 0;
}
```

```
vaibhavi@VYSRI:~/OSSP/src$ nano complex.c
vaibhavi@VYSRI:~/OSSP/src$ gcc complex.c -o ../bin/complex
vaibhavi@VYSRI:~/OSSP/src$ ../bin/complex
myshell>  ls | wc -l
23
vaibhavi@VYSRI:~/OSSP/src$
```

16.

hasini@DESKTOP-UTCBQ51: ~

```
  GNU nano 8.7.1                    background.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char input[100];

    printf("myshell> ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    int background = 0;
    int len = strlen(input);

    if (len > 0 && input[len - 1] == '&')
    {
        background = 1;
        input[len - 1] = '\0';
    }

    pid_t pid = fork();

    if (pid == 0)
    {
        char *args[10];
        int i = 0;

        char *token = strtok(input, " ");

        while (token != NULL)
        {
            args[i++] = token;
            token = strtok(NULL, " ");
        }

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```




```
vaibhavi@VYSRI:~/OSSP/src$ nano background.c
vaibhavi@VYSRI:~/OSSP/src$ gcc background.c -o ../bin/background
vaibhavi@VYSRI:~/OSSP/src$ ../bin/background
myshell> ls
Process started. PID: 1528
append.c      complex.c  history    memory      pipeline.c  redirect.c
background.c  error.txt  history.c  memory.c    pipes.c     stderr.c
builtin.c     escape.c   input      output.txt  process.c   tokenize.c
builtins2.c   fork.c     input.c    parse.c     quotes.c    waitpath.c
Foreground process completed.
vaibhavi@VYSRI:~/OSSP/src$
```

17.

hasini@DESKTOP-UTCBQ51: ~

GNU nano 8.7.1 jobs.c *

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

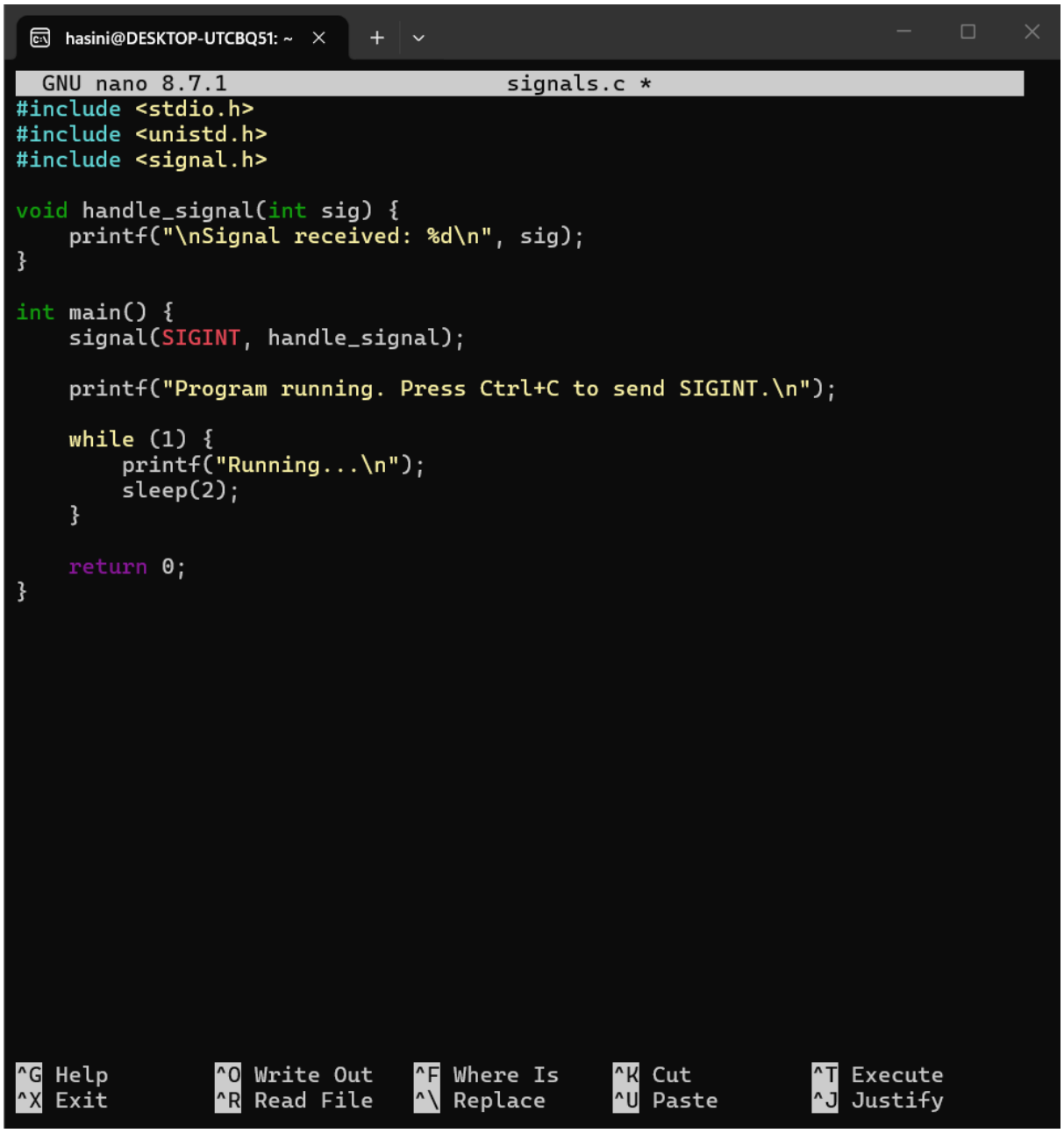
int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child process running. PID: %d\n", getpid());
        sleep(10);
        printf("Child process completed.\n");
    }
    else {
        printf("Parent process: Child PID = %d\n", pid);
        waitpid(pid, NULL, 0);
        printf("Child returned to foreground.\n");
    }

    return 0;
}
```

```
vaibhavi@VYSRI:~/OSSP/src$ nano jobs.c
vaibhavi@VYSRI:~/OSSP/src$ gcc jobs.c -o ../bin/jobs
vaibhavi@VYSRI:~/OSSP/src$ ../bin/jobs
Parent process: Child PID = 1574
Child process running. PID: 1574
Child process completed.
Child returned to foreground.
vaibhavi@VYSRI:~/OSSP/src$
```

18.


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 signals.c *
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
void handle_signal(int sig) {
printf("\nSignal received: %d\n", sig);
}
int main() {
signal(SIGINT, handle_signal);
printf("Program running. Press Ctrl+C to send SIGINT.\n");
while (1) {
printf("Running...\n");
sleep(2);
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

```
vaibhavi@VYSRI:~/OSSP/src$ nano signals.c
vaibhavi@VYSRI:~/OSSP/src$ gcc signals.c -o ../bin/signals
vaibhavi@VYSRI:~/OSSP/src$ ../bin/signals
Program running. Press Ctrl+C to send SIGINT.
Running...
Running...
Running...
Running...
Running...
Running...
^C
Signal received: 2
Running...
Running...
Running...
Running...
Running...
Running...
Running...
Running...
++^HRunning...
Running...
^\Quit                          ../bin/signals
vaibhavi@VYSRI:~/OSSP/src$
```

19.


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 signalhandler.c *
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
void handle_signal(int sig)
{
if (sig == SIGINT)
printf("\nSIGINT received\n");
if (sig == SIGTSTP)
printf("\nSIGTSTP received\n");
}
int main()
{
signal(SIGINT, handle_signal);
signal(SIGTSTP, handle_signal);
printf("Program is running...\n");
printf("Press Ctrl+C or Ctrl+Z to send signals.\n");
while (1)
{
printf("Running...\n");
sleep(2);
}
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

```
vaibhavi@VYSRI:~/OSSP/src$ nano signalhandler.c
vaibhavi@VYSRI:~/OSSP/src$ gcc signalhandler.c -o ../bin/signalhandler
vaibhavi@VYSRI:~/OSSP/src$ ../bin/signalhandler
Program is running...
Press Ctrl+C or Ctrl+Z to send signals.
Running...
Running...
Running...
Running...
Running...
^C
SIGINT received
Running...
Running...
Running...
Running...
Running...
^Z
SIGTSTP received
Running...
Running...
Running...
Running...
^\Quit                          ../bin/signalhandler
vaibhavi@VYSRI:~/OSSP/src$
```

20.

hasini@DESKTOP-UTCBQ51: ~

```
 GNU nano 8.7.1                  processgroup.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

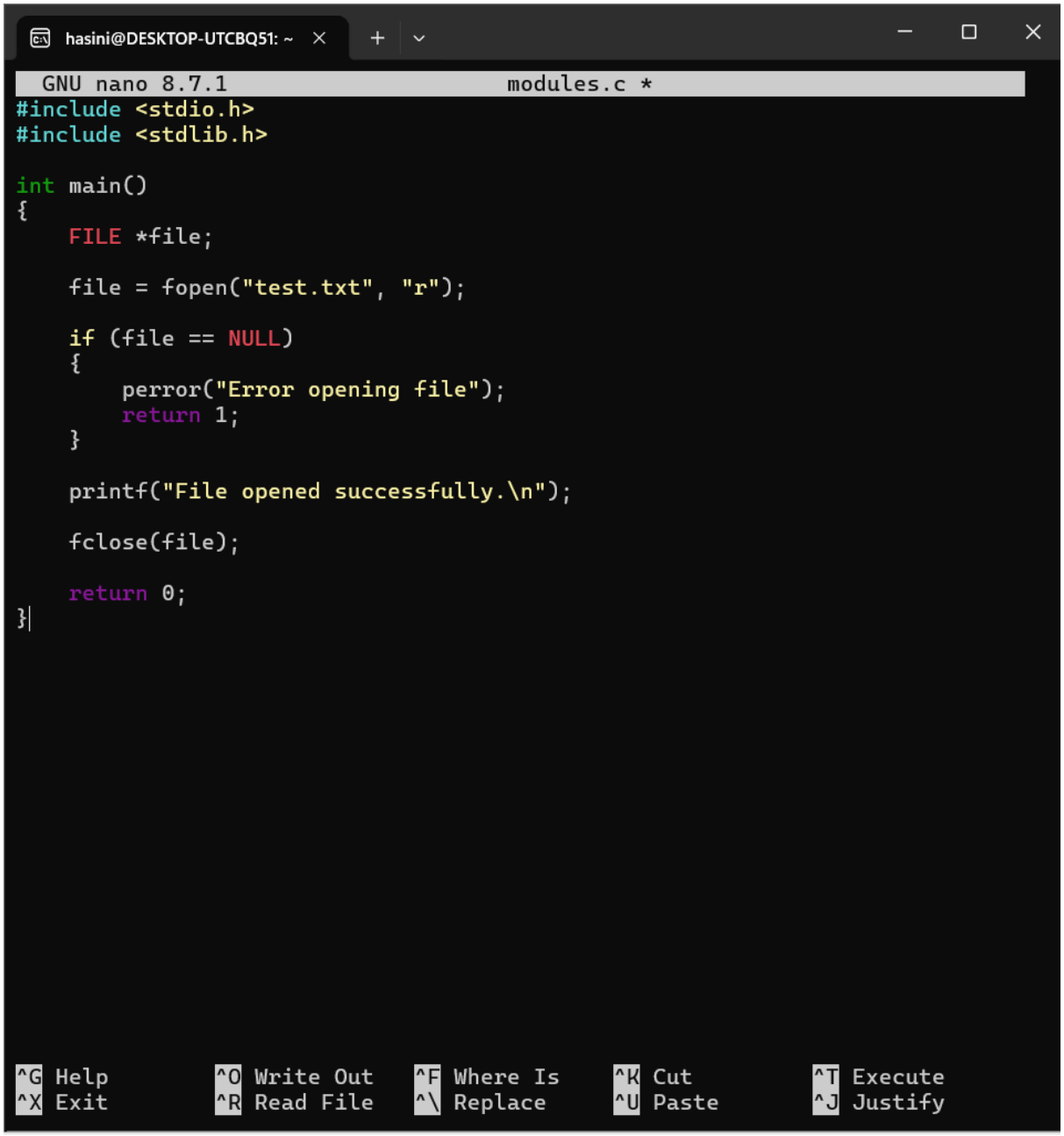
int main()
{
    printf("Process ID: %d\n", getpid());
    printf("Process Group ID: %d\n", getpgrp());
    printf("Parent Process ID: %d\n", getppid());

    return 0;
}
```

```
^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

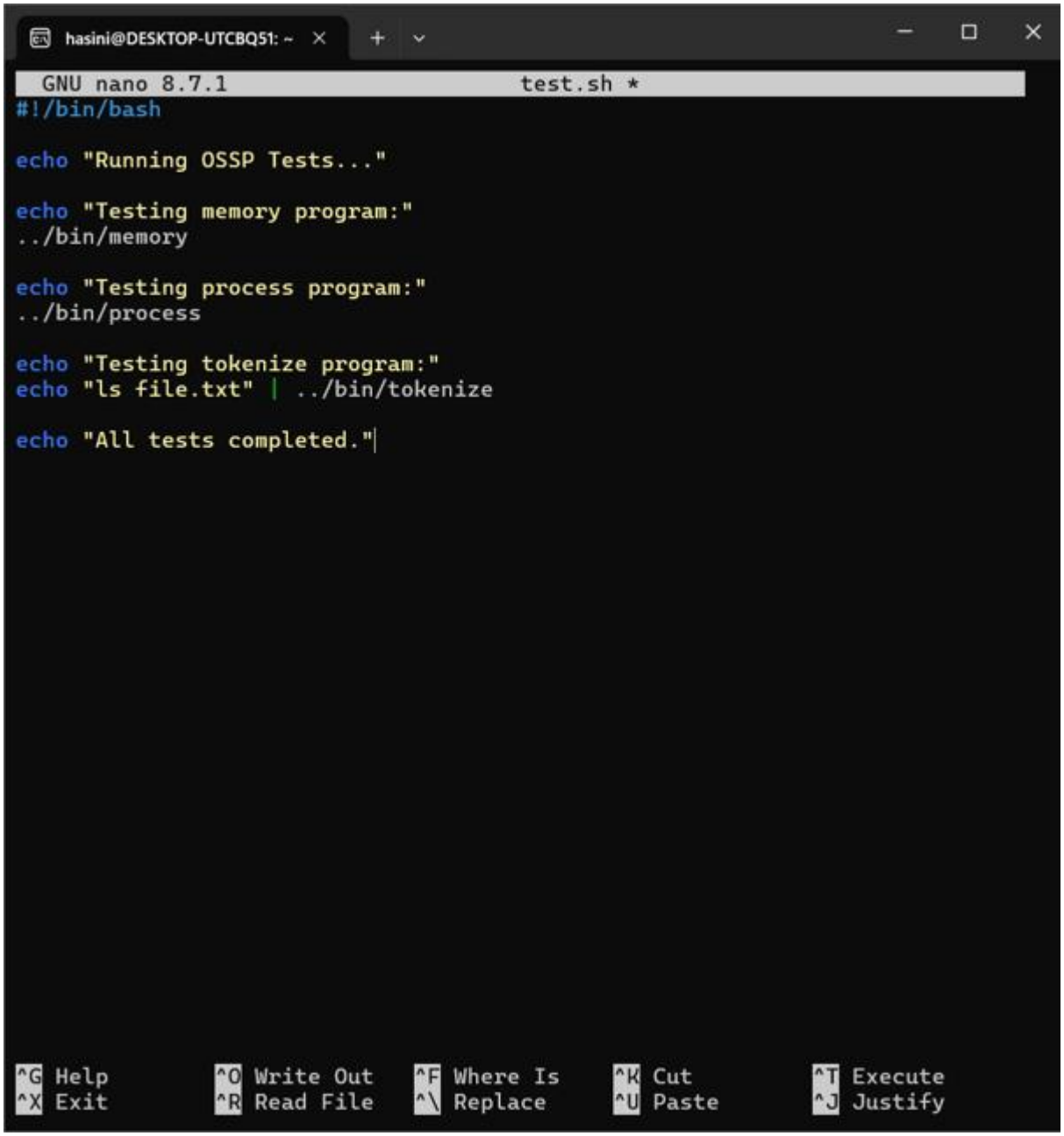
```
vaibhavi@VYSRI:~/OSSP/src$ nano processgroup.c
vaibhavi@VYSRI:~/OSSP/src$ gcc processgroup.c -o ../bin/processgroup
vaibhavi@VYSRI:~/OSSP/src$ ../bin/processgroup
Process ID: 1725
Process Group ID: 1725
Parent Process ID: 329
```

21.




```
vaibhavi@VYSRI:~/OSSP/src$ nano modules.c
vaibhavi@VYSRI:~/OSSP/src$ touch test.txt
vaibhavi@VYSRI:~/OSSP/src$ gcc modules.c -o ../bin/modules
vaibhavi@VYSRI:~/OSSP/src$ ../bin/modules
File opened successfully.
vaibhavi@VYSRI:~/OSSP/src$
```

22. Valgrind and Test Scripts:


hasini@DESKTOP-UTCBQ51: ~
GNU nano 8.7.1 test.sh *
#!/bin/bash
echo "Running OSSP Tests..."
echo "Testing memory program:"
../bin/memory
echo "Testing process program:"
../bin/process
echo "Testing tokenize program:"
echo "ls file.txt" | ../bin/tokenize
echo "All tests completed."
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

```
vaibhavi@VYSRI:~/OSSP/src$ nano test.sh
vaibhavi@VYSRI:~/OSSP/src$ chmod +x test.sh
vaibhavi@VYSRI:~/OSSP/src$ ./test.sh
Running OSSP Tests...
Testing memory program:
./test.sh: line 6: ../bin/memory: No such file or directory
Testing process program:
Process ID: 1819
Parent Process ID: 1817
Testing tokenize program:
Enter a command: Tokens:
ls
file.txt
All tests completed.
vaibhavi@VYSRI:~/OSSP/src$
```

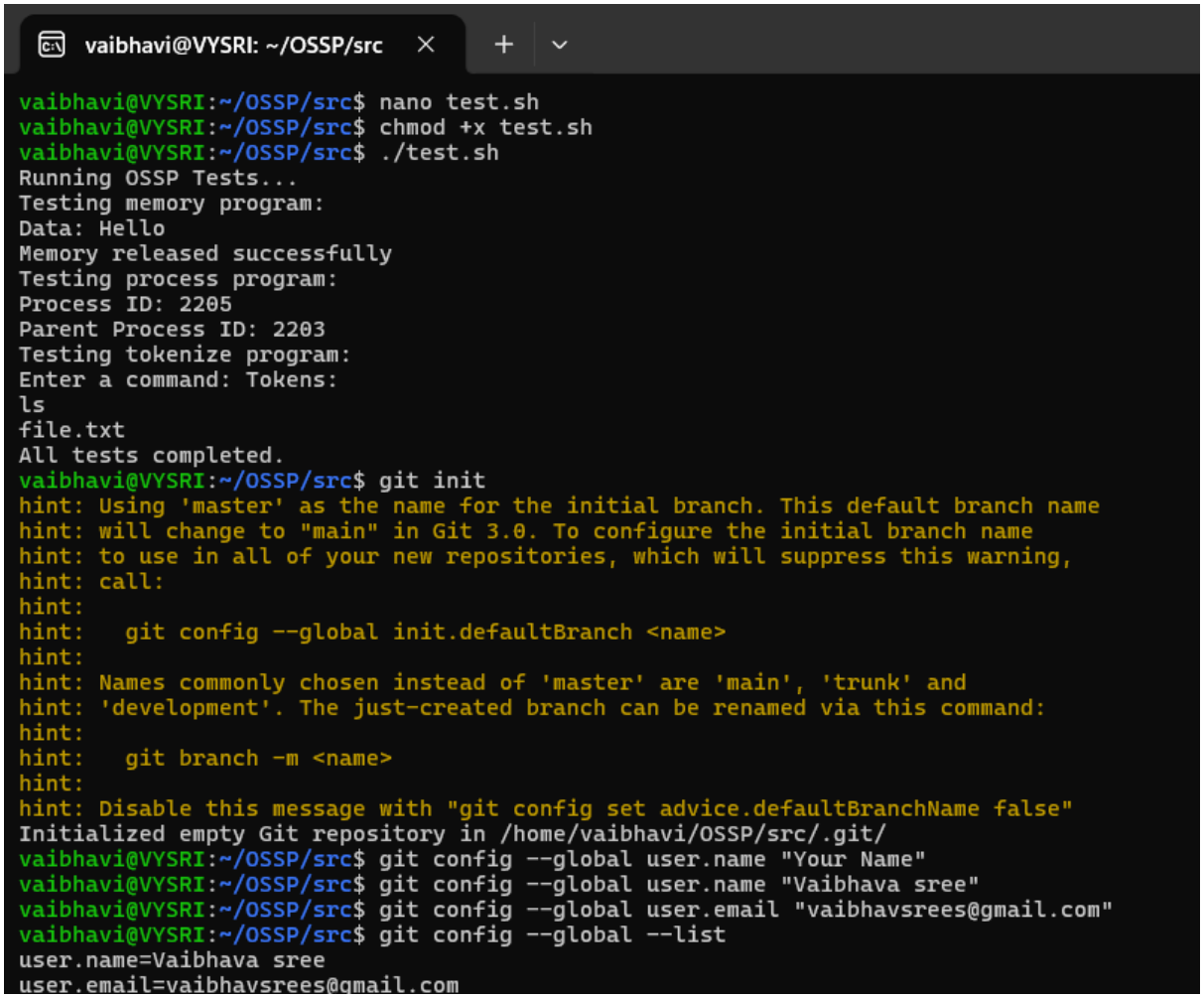
23.

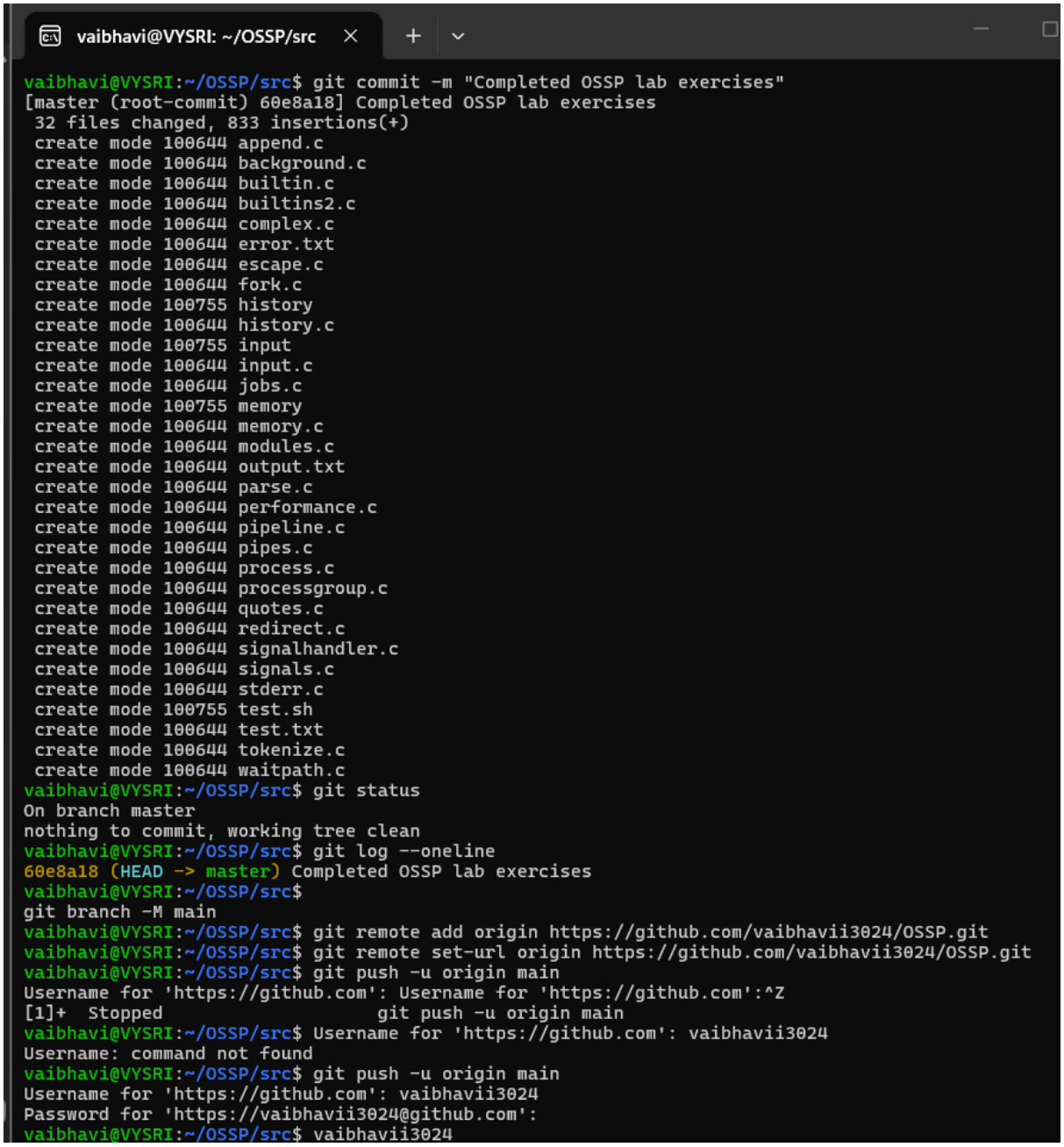



```
vaibhavi@VYSRI:~/OSSP/src$ nano performance.c
vaibhavi@VYSRI:~/OSSP/src$ gcc performance.c -o ../bin/performance
vaibhavi@VYSRI:~/OSSP/src$ ../bin/performance
Execution time: 0.027334 seconds
vaibhavi@VYSRI:~/OSSP/src$
```

24.

```
vaibhavi@VYSRI:~/OSSP/src$ nano memory.c
vaibhavi@VYSRI:~/OSSP/src$ gcc memory.c -o ../bin/memory
vaibhavi@VYSRI:~/OSSP/src$ ./bin/memory
-bash: ./bin/memory: No such file or directory
vaibhavi@VYSRI:~/OSSP/src$ ../bin/memory
Data: Hello
Memory released successfully
vaibhavi@VYSRI:~/OSSP/src$ find . -maxdepth 2 -type f | sort
./append.c
./background.c
./builtin.c
./builtins2.c
./complex.c
./error.txt
./escape.c
./fork.c
./history
./history.c
./input
./input.c
./jobs.c
./memory
./memory.c
./modules.c
./output.txt
./parse.c
./performance.c
./pipeline.c
./pipes.c
./process.c
./processgroup.c
./quotes.c
./redirect.c
./signalhandler.c
./signals.c
./stderr.c
./test.sh
./test.txt
./tokenize.c
./waitpath.c
vaibhavi@VYSRI:~/OSSP/src$
```


vaibhavi@VYSRI: ~/OSSP/src
vaibhavi@VYSRI:~/OSSP/src$ nano test.sh
vaibhavi@VYSRI:~/OSSP/src$ chmod +x test.sh
vaibhavi@VYSRI:~/OSSP/src$ ./test.sh
Running OSSP Tests...
Testing memory program:
Data: Hello
Memory released successfully
Testing process program:
Process ID: 2205
Parent Process ID: 2203
Testing tokenize program:
Enter a command: Tokens:
ls
file.txt
All tests completed.
vaibhavi@VYSRI:~/OSSP/src$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint: git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint: git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/vaibhavi/OSSP/src/.git/
vaibhavi@VYSRI:~/OSSP/src$ git config --global user.name "Your Name"
vaibhavi@VYSRI:~/OSSP/src$ git config --global user.name "Vaibhava sree"
vaibhavi@VYSRI:~/OSSP/src$ git config --global user.email "vaibhavsrees@gmail.com"
vaibhavi@VYSRI:~/OSSP/src$ git config --global --list
user.name=Vaibhava sree
user.email=vaibhavsrees@gmail.com


vaibhavi@VYSRI: ~/OSSP/src
vaibhavi@VYSRI:~/OSSP/src$ git commit -m "Completed OSSP lab exercises"
[master (root-commit) 60e8a18] Completed OSSP lab exercises
32 files changed, 833 insertions(+)
create mode 100644 append.c
create mode 100644 background.c
create mode 100644 builtin.c
create mode 100644 builtins2.c
create mode 100644 complex.c
create mode 100644 error.txt
create mode 100644 escape.c
create mode 100644 fork.c
create mode 100755 history
create mode 100644 history.c
create mode 100755 input
create mode 100644 input.c
create mode 100644 jobs.c
create mode 100755 memory
create mode 100644 memory.c
create mode 100644 modules.c
create mode 100644 output.txt
create mode 100644 parse.c
create mode 100644 performance.c
create mode 100644 pipeline.c
create mode 100644 pipes.c
create mode 100644 process.c
create mode 100644 processgroup.c
create mode 100644 quotes.c
create mode 100644 redirect.c
create mode 100644 signalhandler.c
create mode 100644 signals.c
create mode 100644 stderr.c
create mode 100755 test.sh
create mode 100644 test.txt
create mode 100644 tokenize.c
create mode 100644 waitpath.c
vaibhavi@VYSRI:~/OSSP/src$ git status
On branch master
nothing to commit, working tree clean
vaibhavi@VYSRI:~/OSSP/src$ git log --oneline
60e8a18 (HEAD -> master) Completed OSSP lab exercises
vaibhavi@VYSRI:~/OSSP/src$
git branch -M main
vaibhavi@VYSRI:~/OSSP/src$ git remote add origin https://github.com/vaibhavii3024/OSSP.git
vaibhavi@VYSRI:~/OSSP/src$ git remote set-url origin https://github.com/vaibhavii3024/OSSP.git
vaibhavi@VYSRI:~/OSSP/src$ git push -u origin main
Username for 'https://github.com': Username for 'https://github.com':^Z
[1]+ Stopped git push -u origin main
vaibhavi@VYSRI:~/OSSP/src$ Username for 'https://github.com': vaibhavii3024
Username: command not found
vaibhavi@VYSRI:~/OSSP/src$ git push -u origin main
Username for 'https://github.com': vaibhavii3024
Password for 'https://vaibhavii3024@github.com':
vaibhavi@VYSRI:~/OSSP/src$ vaibhavii3024